

Ark Academy US Pilot Program

Ark Academy US Pilot Program

“The Ark Challenge Files”

- Ages: 8-11
 - Format: Virtual (Google Meet)
 - Group Size: 6 students
 - Duration: 4 sessions
 - Length per Session: 50 minutes
-

OVERALL PROGRAM STRUCTURE

Student Experience Framework

Each session follows this immersive structure:

Phase	Time	Purpose
Mission Briefing	5 mins	Story immersion + challenge setup
Team Investigation	10 mins	Analyze clues and gather evidence
Collaborative Planning	10 mins	Problem-solving + communication
Build / Solve Phase	15 mins	Create solutions using tools
Twist Event	5 mins	Adaptability challenge
Reflection Debrief	5 mins	Connect to Ark values

DIGITAL TOOLS

Tool	Purpose
Google Meet	Main communication hub
Breakout Discussions	Small-group planning
Meet Chat	Secret clues and rapid communication
Polls	Group decisions
Reactions	Silent voting / agreement
Excalidraw	Mapping, brainstorming, systems planning
Google Docs	Shared notes and evidence tracking
Canva	Visual solution creation

SESSION 1 — ESCAPE THE MALL

Theme

Students wake up trapped inside a giant futuristic mall after closing time. The exits are locked, the lights are glitching, and a storm is approaching. To escape, they must solve clues hidden throughout different stores before power completely shuts down.

Learning Goals

Core Values Focus

- Growth Mindset
- Adaptability

Skills

- Team communication
 - Logical deduction
 - Time management
 - Collaborative planning
-

STORY INTRO

A robotic mall assistant named “M.A.RA” appears on screen:

“Emergency lockdown activated. Main exits sealed. Power remaining: 45 minutes.”

Students are recruited as emergency problem-solvers to escape before the building goes dark.

FLOW

1. Mission Briefing (5 mins)

Teacher:

- Changes virtual background to mall security room
- Shares countdown timer
- Plays alarm sound effects

Students receive:

- Mall map
 - First clue card
-

2. Team Investigation (10 mins)

Students examine:

- store receipts
- locker codes
- hidden symbols
- timestamps

Teacher shares clue cards through screen share.

Students use:

- Google Doc to organize clues
 - Meet chat for quick theories
-

3. Collaborative Planning (10 mins)

Students decide:

- Which “store” to investigate next
- What clues connect together

Teacher intentionally gives misleading clues to encourage persistence.

4. Escape Build Challenge (15 mins)

Students create:

- An escape route map in Excalidraw
 - Final code sequence
 - Emergency escape explanation
-

5. Twist Event (5 mins)

Teacher announces:

“Power outage in Sector 3. Half your clues are now inaccessible.”

Students must adapt their strategy quickly.

6. Reflection Debrief (5 mins)

Discussion prompts:

- What mistake helped your team learn?
 - When did you adapt your thinking?
 - How did your team stay calm?
-

SESSION 2 — CONUNDRUM: THE FOUR-SEASON ISLANDS

Theme

Three islands must be connected using transportation systems. However, each island experiences a completely different season year-round:

- Winter Island
- Summer Island
- Storm Island

Students must design safe and sustainable connection paths for the citizens.

Learning Goals

Core Values Focus

- Empathy
- Adaptability

Skills

- Systems thinking
 - Engineering creativity
 - Community-centered design
 - Perspective-taking
-

STORY INTRO

The mayor appears requesting urgent help:

“Families cannot visit one another anymore. Supplies are delayed. We need your help to reconnect the islands.”

FLOW

1. Mission Briefing (5 mins)

Teacher presents:

- island maps
- weather conditions
- citizen profiles

Example:

- elderly citizens
 - children
 - farmers
 - delivery workers
-

2. Investigation & Research (10 mins)

Students analyze:

- terrain
- weather hazards
- transportation limits

Teacher introduces:

- budget limitations
 - environmental restrictions
-

3. Collaborative Planning (10 mins)

Students brainstorm:

- bridges
- tunnels
- cable systems
- seasonal vehicles

Using Excalidraw:

- sketch island connections
-

4. Design Challenge (15 mins)

Students create:

- transportation proposal in Canva
 - connection system blueprint
 - emergency weather adaptation plan
-

5. Twist Event (5 mins)

Teacher reveals:

"A blizzard has started on Winter Island."

Students must redesign routes in real-time.

6. Reflection Debrief (5 mins)

Questions:

- Who did your design help most?
 - How did your group handle disagreements?
 - What changed after the blizzard twist?
-

SESSION 3 — MYSTERY OF THE MISSING STUDENT

Theme

A student named Eli disappeared after school. Students must analyze test papers, notes, schedules, and coded clues to determine Eli's location before nightfall.

Learning Goals

Core Values Focus

- Empathy
- Resilience

Skills

- Deductive reasoning
- Observation
- Communication
- Emotional awareness

STORY INTRO

Teacher acts as school principal:

“Eli never made it home. We found strange markings on his papers.”

Students become investigation teams.

FLOW

1. Mission Briefing (5 mins)

Teacher shares:

- scanned “test papers”
 - attendance logs
 - notebook pages
-

2. Evidence Analysis (10 mins)

Students search for:

- hidden codes
- repeated symbols
- map references
- acrostics

Google Docs becomes:

- shared detective notebook
-

3. Investigation Planning (10 mins)

Students divide roles:

- code breaker
- mapper
- recorder
- clue connector

Teacher sends secret clues privately through Meet chat.

4. Final Deduction Challenge (15 mins)

Students determine:

- Eli's location
- why Eli disappeared
- safest rescue plan

Students present:

- evidence board
 - reasoning explanation
-

5. Twist Event (5 mins)

Teacher reveals:

"One clue was intentionally fake."

Students must reassess evidence.

6. Reflection Debrief (5 mins)

Questions:

- How did your team react to being wrong?
 - What helped your team keep going?
 - How did empathy help solve the mystery?
-

SESSION 4 — NO ONE GETS LEFT BEHIND

Theme

Earth's atmosphere is collapsing. Students must design a rocket ship evacuation system using limited materials while ensuring every passenger can safely fit onboard.

The central rule:

Nobody can be excluded.

Learning Goals

Core Values Focus

- Empathy
- Resilience
- Growth Mindset

Skills

- Resource management
 - Ethical problem-solving
 - Engineering collaboration
 - Negotiation
-

STORY INTRO

Students receive a transmission:

"The evacuation launch begins in 40 minutes. Materials are critically low."

FLOW

1. Mission Briefing (5 mins)

Teacher presents:

- passenger list
- raw materials
- spaceship constraints

Example constraints:

- oxygen limits
 - weight restrictions
 - damaged engines
-

2. Resource Analysis (10 mins)

Students evaluate:

- what materials to prioritize
- how to maximize space
- passenger needs

Special challenge:

Some passengers have unique needs:

- elderly
- injured

- children
 - pets
-

3. Collaborative Engineering (10 mins)

Students sketch rocket layouts in Excalidraw.

Must negotiate:

- sleeping areas
 - oxygen systems
 - food storage
 - seating
-

4. Rocket Design Challenge (15 mins)

Students create:

- Canva spaceship blueprint
 - evacuation explanation
 - survival strategy
-

5. Twist Event (5 mins)

Teacher announces:

“Two additional survivors arrived unexpectedly.”

Students must redesign quickly.

6. Reflection Debrief (5 mins)

Questions:

- What sacrifices did your team make?
 - How did you ensure fairness?
 - What helped your group persist?
-

FACILITATOR ROLE

The facilitator acts as:

- narrator
- game master
- clue controller
- emotional coach

The teacher should:

- encourage student-led thinking
 - avoid giving direct answers
 - reward creative attempts
 - highlight collaboration moments
-

ASSESSMENT STYLE

Ark's assessment model should focus on observable behaviors rather than right/wrong answers.

Core Value	Observable Indicators
Growth Mindset	tries new strategies after failure
Resilience	continues despite confusion
Empathy	considers others in decisions

Adaptability	adjusts plan after new information
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Objectives

ARK ACADEMY US PILOT PROGRAM

PROGRESSIVE SESSION OBJECTIVES

Focused on the 4 Core Values:

- Growth Mindset
- Resilience
- Empathy
- Adaptability

The progression of objectives is intentionally designed to move students from:

1. Individual participation and exploration
2. Collaborative awareness
3. Complex reasoning under uncertainty
4. Ethical and inclusive problem-solving leadership

Each session increases in:

- emotional complexity
 - collaborative demand
 - strategic thinking
 - decision-making responsibility
-

SESSION 1 — ESCAPE THE MALL

Core Focus:

Growth Mindset + Adaptability

Developmental Stage:

Exploration and Flexible Thinking

Session Objective

Students will collaboratively investigate and solve a time-sensitive mystery challenge by experimenting with multiple strategies, adjusting to unexpected obstacles, and communicating ideas effectively within a team environment.

Core Value Objectives

Growth Mindset Objective

Students will demonstrate willingness to try different approaches, learn from unsuccessful attempts, and remain engaged during difficult problem-solving moments.

Adaptability Objective

Students will respond constructively to changing information and unexpected twists by revising plans and adjusting team strategies.

Collaboration Objective

Students will practice contributing ideas, listening to peers, and participating actively within a shared virtual investigation space.

Success Indicators

By the end of the session, students should be able to:

- explain how their team changed strategies during the mission
- identify at least one mistake that helped improve their thinking
- contribute meaningfully to collaborative planning discussions
- remain engaged despite uncertainty or incorrect assumptions

SESSION 2 — CONUNDRUM: THE FOUR-SEASON ISLANDS

Core Focus:

Empathy + Adaptability

Developmental Stage:

Perspective-Taking and Systems Thinking

Session Objective

Students will collaboratively design transportation and connection systems that address environmental, social, and community challenges while considering the diverse needs of different groups of people.

Core Value Objectives**Empathy Objective**

Students will demonstrate awareness of how decisions impact different community members and incorporate inclusive thinking into group solutions.

Adaptability Objective

Students will revise plans effectively when environmental conditions and limitations change unexpectedly.

Growth Mindset Objective

Students will remain open to improving and redesigning ideas after receiving new information or feedback.

Success Indicators

By the end of the session, students should be able to:

- explain how their design supported multiple types of people
- demonstrate flexibility when redesigning under changing conditions
- contribute solutions that balance creativity and practicality
- collaborate respectfully during disagreements or differing viewpoints

SESSION 3 — MYSTERY OF THE MISSING STUDENT

Core Focus:

Resilience + Empathy

Developmental Stage:

Reasoning Through Complexity and Emotional Awareness

Session Objective

Students will analyze clues, evaluate evidence, and collaboratively solve a mystery while demonstrating perseverance, emotional regulation, and thoughtful teamwork during moments of uncertainty.

Core Value Objectives**Resilience Objective**

Students will continue contributing productively despite misleading clues, confusion, or unsuccessful attempts.

Empathy Objective

Students will demonstrate perspective-taking and emotional awareness while investigating the circumstances surrounding the missing student.

Adaptability Objective

Students will reassess assumptions and revise conclusions when presented with contradictory evidence.

Success Indicators

By the end of the session, students should be able to:

- explain how their team recovered after incorrect assumptions
- demonstrate calm and productive participation during setbacks
- analyze evidence logically before making conclusions
- communicate respectfully while solving disagreements within the team

SESSION 4 — NO ONE GETS LEFT BEHIND

Core Focus:

Empathy + Resilience + Leadership Adaptability

Developmental Stage:

Ethical Decision-Making and Inclusive Leadership

Session Objective

Students will collaboratively design a survival solution under limited resources while ensuring fairness, inclusion, and responsible decision-making for all members of the community.

Core Value Objectives

Empathy Objective

Students will prioritize inclusive decision-making by considering the needs, safety, and wellbeing of all passengers during resource allocation and design planning.

Resilience Objective

Students will maintain collaboration and constructive problem-solving despite increasing pressure, constraints, and unexpected complications.

Adaptability Objective

Students will revise systems and negotiate solutions effectively when new challenges and limitations emerge.

Growth Mindset Objective

Students will demonstrate openness to improving ideas collaboratively and refining solutions through discussion and feedback.

Success Indicators

By the end of the session, students should be able to:

- explain how fairness influenced their final decisions
 - demonstrate collaborative problem-solving under pressure
 - revise plans effectively after unexpected changes
 - justify decisions using logical and ethical reasoning
 - ensure all team members had opportunities to contribute
-

OVERALL PROGRAM OBJECTIVE

By the end of the four-session pilot program, students will demonstrate increasing ability to:

- collaborate effectively during complex team-based challenges
- adapt strategies during uncertain and changing situations
- persist through setbacks using resilient problem-solving behaviors
- demonstrate empathy and inclusion during group decision-making
- communicate ideas clearly within virtual collaborative environments
- apply creative and critical thinking to real-world inspired problems

Students will progress from:

- solving guided collaborative mysteries
to
 - managing complex ethical and systems-based challenges that require leadership, adaptability, and inclusive thinking.
-

PROGRESSION OF CORE VALUE DEVELOPMENT

Session	Growth Focus	Student Development Progression
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Session 1	Adaptability & Exploration	Learning to adjust and participate actively
Session 2	Perspective-Taking	Understanding how solutions affect others
Session 3	Resilience Under Uncertainty	Managing setbacks and conflicting information
Session 4	Ethical Leadership & Inclusion	Making responsible decisions for the whole group

ARK ACADEMY PILOT LEARNING OUTCOME

The pilot experience is designed to help students develop:

- collaborative intelligence
- emotional-social awareness
- adaptive thinking
- resilience under uncertainty
- ethical problem-solving
- creative confidence

through immersive, mission-based experiences rather than traditional lecture-based instruction.

ASSESSMENT FRAMEWORK

ASSESSMENT FRAMEWORK, RUBRICS & SELF-EVALUATION SYSTEM

PROGRAM PHILOSOPHY

Ark Academy's assessment model evaluates how students think, collaborate, adapt, and respond to complex situations rather than measuring memorization or traditional academic performance alone.

Students are assessed through:

- observable behaviors
- collaboration patterns
- communication effectiveness
- emotional-social responses
- problem-solving processes
- adaptability during uncertainty

Assessment occurs continuously during immersive mission-based experiences.

PERFORMANCE SCALE

Score	Level	Description
4	Advanced Demonstration	Consistently demonstrates the skill independently and positively influences group performance
3	Proficient Demonstration	Frequently demonstrates the skill with minimal facilitator support
2	Developing Demonstration	Demonstrates the skill inconsistently and requires occasional support or prompting
1	Emerging Demonstration	Beginning to develop the skill and requires frequent support and guidance

CORE ASSESSMENT DOMAINS

1. GROWTH MINDSET

Definition

The ability to embrace challenges, learn through mistakes, remain open to feedback, and persist through uncertainty.

Observable Indicators

- Attempts multiple strategies before giving up
- Responds positively to corrections or feedback
- Takes intellectual risks during challenges
- Demonstrates curiosity and willingness to experiment
- Learns from unsuccessful attempts
- Re-engages after setbacks

Rubric

Score	Indicators
4	Independently experiments with multiple strategies, reflects on mistakes constructively, and encourages teammates to persist through challenges
3	Demonstrates willingness to retry tasks and adapt strategies after setbacks with minimal support
2	Attempts new approaches inconsistently and may become discouraged without facilitator encouragement
1	Avoids difficult tasks, gives up quickly, or resists feedback and alternative strategies

2. RESILIENCE

Definition

The ability to maintain emotional regulation, persistence, and productive engagement during stressful or difficult situations.

Observable Indicators

- Remains engaged during difficult moments
- Continues participating after confusion or failure
- Demonstrates emotional control under pressure
- Recovers productively from setbacks
- Maintains effort despite time constraints

Rubric

Score	Indicators
4	Consistently remains calm, solution-focused, and emotionally regulated during complex or high-pressure situations
3	Demonstrates persistence and maintains participation during most challenging moments
2	Occasionally disengages or becomes frustrated but can recover with support
1	Frequently withdraws, shuts down, or struggles to continue participation during difficulty

3. EMPATHY & SOCIAL AWARENESS

Definition

The ability to recognize, consider, and respond thoughtfully to the perspectives, emotions, and needs of others.

Observable Indicators

- Listens respectfully to teammates
- Includes quieter students in discussions
- Demonstrates awareness of group needs
- Responds kindly during disagreements
- Considers fairness and inclusion during decision-making
- Supports peers emotionally and academically

Rubric

Score	Indicators
4	Actively promotes inclusion, demonstrates strong perspective-taking, and supports group cohesion consistently
3	Collaborates respectfully and considers others' ideas during discussions and decision-making
2	Participates appropriately but inconsistently demonstrates awareness of others' perspectives
1	Frequently interrupts, dismisses peers, or struggles to collaborate respectfully

4. ADAPTABILITY

Definition

The ability to adjust thinking, strategies, and behavior effectively when conditions, information, or expectations change.

Observable Indicators

- Adjusts plans after receiving new information
- Demonstrates flexibility during unexpected twists
- Responds constructively to changing constraints
- Revises ideas efficiently
- Maintains productivity during uncertainty

Rubric

Score

Indicators

- | | |
|---|---|
| 4 | Quickly adapts strategies independently and helps guide teammates through changing situations |
| 3 | Adjusts plans appropriately after new information or unexpected changes |
| 2 | Demonstrates some flexibility but may struggle to shift strategies efficiently |
| 1 | Resists change or struggles significantly when plans must be modified |
-

COLLABORATION & COMMUNICATION

5. COLLABORATIVE COMMUNICATION

Definition

The ability to express ideas clearly, listen actively, and contribute meaningfully within a team setting.

Observable Indicators

- Shares ideas clearly and respectfully
- Participates actively in discussions
- Builds upon others' ideas
- Uses collaborative language
- Helps maintain productive discussions
- Uses digital collaboration tools effectively

Rubric

Score

Indicators

- 4 Communicates clearly, facilitates productive discussion, and consistently strengthens team collaboration
 - 3 Shares ideas appropriately and contributes positively to group communication
 - 2 Participates inconsistently or requires prompting to engage in discussions
 - 1 Rarely contributes or struggles to communicate ideas respectfully and effectively
-

6. CRITICAL THINKING & PROBLEM SOLVING

Definition

The ability to analyze information, identify patterns, evaluate solutions, and make reasoned decisions during complex tasks.

Observable Indicators

- Identifies logical connections between clues
- Evaluates multiple possible solutions
- Analyzes evidence carefully
- Questions assumptions thoughtfully
- Demonstrates reasoning during decision-making
- Applies prior knowledge to new situations

Rubric

Score	Indicators
4	Demonstrates sophisticated reasoning, identifies patterns independently, and contributes high-level analytical insights
3	Uses logical reasoning effectively to support group problem-solving
2	Demonstrates emerging analytical thinking but requires occasional guidance

- 1 Struggles to connect information logically or contribute to solution-building
-

7. CREATIVE & INNOVATIVE THINKING

Definition

The ability to generate original ideas, explore possibilities, and design imaginative solutions within constraints.

Observable Indicators

- Suggests creative approaches
- Combines ideas innovatively
- Thinks beyond obvious solutions
- Demonstrates imagination during design tasks
- Explores unconventional possibilities

Rubric

Score	Indicators
4	Frequently contributes highly original ideas that significantly enhance team solutions
3	Demonstrates creativity and contributes useful innovative ideas during challenges
2	Occasionally contributes creative ideas but tends to rely on familiar approaches
1	Rarely contributes original ideas or struggles with open-ended thinking

DIGITAL COLLABORATION & TOOL FLUENCY

8. DIGITAL COLLABORATION

Definition

The ability to participate responsibly and effectively within virtual collaborative environments using digital tools.

Observable Indicators

- Navigates collaborative tools independently
- Contributes to shared digital workspaces
- Uses Meet features appropriately
- Demonstrates online collaboration etiquette
- Engages consistently during virtual sessions

Rubric

Score	Indicators
4	Uses digital tools independently and enhances group productivity through effective virtual collaboration
3	Participates appropriately and uses digital tools effectively with minimal support
2	Demonstrates developing digital collaboration skills but occasionally requires guidance
1	Struggles to engage productively within virtual collaborative environments

SESSION-SPECIFIC ASSESSMENT FOCUS

Session	Primary Skill Focus
Escape the Mall	Adaptability, Logical Reasoning, Team Communication
Conundrum: Four-Season Islands	Systems Thinking, Empathy, Collaborative Planning

Mystery of the Missing Student Deductive Reasoning, Resilience, Evidence Analysis

No One Gets Left Behind Ethical Decision-Making, Resource Management,
Inclusive Collaboration

FACILITATOR OBSERVATION NOTES

Facilitators should collect evidence through:

- direct observation
- student dialogue
- decision-making behaviors
- collaboration patterns
- responses to setbacks
- quality of reasoning
- participation within shared tools

Notes should focus on:

- growth trends
 - behavioral consistency
 - emerging strengths
 - support needs
 - collaboration dynamics
-

STUDENT SELF-EVALUATION EXIT TICKET

To be completed at the end of every session.

Reflection Questions

Growth Mindset

1. What challenge was hardest for you today?
2. What strategy did you try when something did not work?

Collaboration

3. How did you help your team today?
4. How did your teammates help you?

Adaptability

5. What changed during the mission that forced your team to adjust?
6. How did you react when the plan changed?

Emotional Reflection

7. What moment made you feel proud today?
8. What is one thing you would improve next time?

Self-Rating Scale

Students rate themselves from 1-4:

Skill	Self Score
I contributed ideas to my team	
I listened to others respectfully	
I stayed focused during challenges	
I kept trying even when things were difficult	
I adapted when plans changed	

FINAL REPORT CARD FORMAT

Student Strengths

Narrative feedback highlighting:

- strongest competencies
- collaboration behaviors
- problem-solving approaches
- leadership moments

Growth Areas

Specific developmental next steps tied to:

- communication
- flexibility
- resilience
- reasoning
- teamwork

Skill Radar Overview

Domains:

- Growth Mindset
- Resilience
- Empathy
- Adaptability
- Collaboration
- Critical Thinking
- Creativity
- Digital Collaboration

Facilitator Summary

Overall developmental observations based on all four missions.

ASSESSMENT PHILOSOPHY SUMMARY

Ark Academy's assessment system prioritizes:

- real-world competency development
- collaborative intelligence
- emotional-social learning
- adaptive problem solving
- future-ready skills
- reflective growth

The goal is not simply to measure whether students solved the challenge, but to understand how they approached uncertainty, collaborated with others, and developed through the process.

Canva Instructions for Grace

Ark Academy US Pilot Program

Session 1: ESCAPE THE MALL

The Ark Challenge Files

Ages: 8–11

Duration: 50 Minutes

Students: 6

Core Values: Growth Mindset + Adaptability

Supporting Skills: Communication, Collaboration, Deduction, Planning

SESSION OVERVIEW

Big Idea

Students become emergency response agents trapped inside a futuristic smart mall after a system-wide malfunction. Through teamwork, communication, and flexible thinking, they must uncover clues, solve puzzles, adapt to unexpected challenges, and escape before the building loses power.

FACILITATOR SETUP

Virtual Background

Futuristic Mall Security Control Room

Sound Effects

- Alarm siren
- Security announcement
- Power outage sound
- Countdown timer ticking

Materials Needed

- Google Meet
 - Google Docs Evidence Tracker
 - Excalidraw Board
 - Canva Template (Escape Route Map)
 - Clue Cards Slides
-

CANVA STORYBOARD FLOW FOR DESIGNER

SLIDE 1

Welcome Screen

THE ARK CHALLENGE FILES

Mission 001: Escape The Mall

Visual:

- Dark futuristic mall
- Flashing emergency lights
- Closed security gates
- Countdown timer graphic

Text:

Emergency Lockdown Activated

Power Remaining: 45 Minutes

Can Your Team Escape?

SLIDE 2

Meet M.A.RA

Visual:

Friendly futuristic robot assistant

Dialogue:

Greetings Investigators.

I am M.A.RA.

Mall Assistance and Rescue Android.

A severe storm has damaged our systems.

All exits have been sealed.

Power reserves are rapidly decreasing.

You have been selected as emergency problem-solvers.

Find the escape code before total shutdown.

SLIDE 3

Mission Objectives

Visual:

Mission briefing dashboard

Students Must:

- ✓ Investigate clues
 - ✓ Communicate effectively
 - ✓ Adapt when plans fail
 - ✓ Unlock the main exit
 - ✓ Escape before power loss
-

PHASE 1

Mission Briefing (5 Minutes)

SLIDE 4

Mall Map

Show:

 Tech Store

 Food Court

 Bookstore

 Arcade

 Cinema

 Main Exit

Teacher Script:

Somewhere inside these stores are clues that reveal the final escape code.

You must work together to decide where to investigate first.

SLIDE 5

First Clue

Visual:

Crumpled receipt

Receipt:

TECH STORE

Item #1 = 7

Item #2 = ?

Time: 7:14 PM

Message written on back:

The first number hides where knowledge sleeps.

Students discuss:

"What place could that be?"

Expected answer:

Bookstore

PHASE 2

Team Investigation (10 Minutes)

SLIDE 6

Investigation Station 1

The Bookstore

Visual:

Bookshelves

Clue Found:

Book #14

Title:

"THE KEY TO EXITS"

Highlighted letters:

K _ _ _

Teacher Prompt:

What letters might complete the word?

Students infer:

KEYS

Hidden Number:

4

SLIDE 7

Investigation Station 2

Arcade

Visual:

Broken arcade machine

Message:

HIGH SCORE

7214

Hint:

Reverse what you think you know.

Students discover:

4127

Hidden Number:

1

SLIDE 8

Investigation Station 3

Food Court

Visual:

Digital menu board

Challenge:

Pizza = 3

Burger = 5

Taco = 2

What is:

Pizza + Taco?

Answer:

5

Hidden Number:

5

SLIDE 9

Investigation Station 4

Cinema

Visual:

Movie poster

Poster says:

SHOWTIME

7:15 PM

8:15 PM

9:15 PM

Question:

What pattern do you notice?

Answer:

+1 hour

Hidden Number:

6

Evidence Tracker

Google Doc Sections:

Clue Source	What We Found	Possible Meaning
Bookstore		
Arcade		
Food Court		
Cinema		

Students record ideas.

PHASE 3

Collaborative Planning (10 Minutes)

SLIDE 10

Team Strategy Room

Prompt:

You have found:

4

1

5

6

But what is the correct order?

Teacher introduces discussion.

Students must explain:

- Why a clue matters
 - What evidence supports it
 - Which clue should come first
-

SLIDE 11

Misleading Evidence

Teacher reveals:

Security Log:

EXIT CODE:

6 _ _ _

Teacher Script:

Are we sure this information is trustworthy?

Purpose:

Introduce uncertainty.

Growth Mindset Focus:

Students learn that not every clue is useful.

Sometimes good investigators must test ideas.

PHASE 4

Escape Build Challenge (15 Minutes)

SLIDE 12

Build Your Escape Plan

Students use Excalidraw.

Task:

Draw:

- Investigation route
- Clue locations
- Final escape path

Must explain:

Why they chose each location.

SLIDE 13

Final Escape Code Puzzle

Hidden message:

BOOKSTORE → ARCADE → FOOD COURT → CINEMA

Code:

4 - 1 - 5 - 6

Students submit answer.

SLIDE 14

Emergency Briefing

Prompt:

Explain:

How did your team solve the mystery?

Sentence starters:

- We first thought...
- Then we realized...
- Our evidence showed...
- Therefore...

PHASE 5

Twist Event (5 Minutes)

SLIDE 15

POWER FAILURE!

Visual:

Red flashing screen

Announcement:

ALERT!

Sector 3 has lost power.

All Food Court records have disappeared.

Remove clue #5.

Students now must reconstruct missing information from memory and notes.

SLIDE 16

Adapt Fast

Teacher Prompt:

Your evidence is incomplete.

How can your team continue?

Expected behaviors:

- Review notes
- Listen to teammates
- Reconstruct information
- Adjust strategy

Adaptability Discussion:

Good problem-solvers change plans when circumstances change.

PHASE 6

Reflection Debrief (5 Minutes)

SLIDE 17

Ark Reflection Circle

Discussion Questions

Growth Mindset

1. Which clue was hardest to solve?
 2. What did you do when your first idea was wrong?
 3. What mistake helped your team move forward?
-

Adaptability

1. How did you react when clues disappeared?
 2. What changed in your team's plan?
 3. What would you do differently next time?
-

Collaboration

1. Who helped you think differently?
 2. How did your team make decisions?
 3. What communication strategy worked best?
-

SLIDE 18

Mission Complete

Visual:

Mall doors opening

M.A.RA Message:

Congratulations Investigators.

You successfully escaped the mall.

More importantly...

You demonstrated the skills of an Ark Explorer:

Growth Mindset

Adaptability

Teamwork

Mission Status:

SUCCESS

FACILITATOR ASSESSMENT RUBRIC

Skill	Emerging	Developing	Proficient	Advanced
Growth Mindset	Gives up when stuck	Tries again with support	Persists through challenge	Actively seeks new strategies
Adaptability	Struggles with changes	Adjusts after prompting	Revises plans independently	Helps team adapt quickly
Collaboration	Rarely contributes	Participates occasionally	Shares ideas consistently	Builds on others' ideas effectively
Communication	Limited explanation	Basic explanations	Clear reasoning and listening	Facilitates productive discussion

Exit Ticket (1 Minute)

Type in chat:

One clue that challenged me was...

One way I adapted today was...

One teammate who helped me was...

My confidence level solving mysteries now is (1–5): ★★★★★

This structure turns the session into a true immersive Ark experience where students repeatedly practice growth mindset and adaptability through meaningful challenge, rather than simply hearing about those values.

Ark Academy US Pilot Program

Session 2: CONUNDRUM – THE FOUR-SEASON ISLANDS

The Ark Challenge Files

Ages: 8–11

Duration: 50 Minutes

Students: 6

Core Values: Empathy + Adaptability

Supporting Skills: Systems Thinking, Engineering Design, Communication, Perspective-Taking

SESSION OVERVIEW

Big Idea

Students become community design consultants tasked with solving a major transportation crisis. Three islands experiencing completely different weather conditions have become disconnected, leaving families separated and supplies delayed. Students must design transportation systems that consider the needs of all citizens while adapting to environmental challenges.

Unlike Session 1's mystery-solving focus, this session shifts students into systems thinking and human-centered design, requiring them to think beyond themselves and consider how solutions affect diverse groups of people.

FACILITATOR SETUP

Virtual Background

World Connection Council Headquarters

Sound Effects

- Ocean waves
- Emergency broadcast
- Wind and storm sounds
- Helicopter flyover sound
- News report transitions

Materials Needed

- Google Meet
 - Excalidraw Design Board
 - Google Docs Planning Sheet
 - Canva Transportation Proposal Template
 - Island Map Slides
 - Citizen Profile Cards
 - Weather Event Cards
-

LEARNING TARGETS

Students Will:

- Design transportation systems that help multiple groups of people
 - Consider how different citizens experience challenges differently
 - Adapt plans when environmental conditions change
 - Practice respectful collaboration during complex decision-making
 - Balance creativity with practicality
-

CANVA STORYBOARD FLOW FOR DESIGNER

SLIDE 1

Welcome Screen

THE ARK CHALLENGE FILES

Mission 002:

CONUNDRUM: THE FOUR-SEASON ISLANDS

Visual:

Three floating islands connected by broken bridges

One island covered in snow

One island covered in sunshine

One island surrounded by lightning storms

Text:

Families are separated.

Supplies cannot arrive.

The islands need your help.

SLIDE 2

Emergency News Report

Visual:

Breaking News screen

Teacher reads dramatically:

This is an emergency broadcast.

Transportation systems connecting the Four-Season Islands have collapsed.

Citizens can no longer travel safely.

Food deliveries are delayed.

Medical supplies are running low.

Families are stranded.

Community engineers are urgently needed.

SLIDE 3

Meet Mayor Aurora

Visual:

Mayor standing beside a large map

Dialogue:

Hello Ark Problem Solvers.

Our islands have always relied on one another.

Winter Island grows medicine.

Summer Island grows food.

Storm Island generates electricity.

Without transportation, our communities cannot survive.

We need your help reconnecting our world.

PHASE 1

Mission Briefing (5 Minutes)

SLIDE 4

The Three Islands

Visual Map

Winter Island ❄️

- Snow all year
 - Slippery terrain
 - Strong winds
 - Produces medicine
-

Summer Island ☀️

- Hot temperatures
 - Large farms
 - Busy marketplaces
 - Produces food
-

Storm Island ⛅️

- Heavy rain
 - Frequent lightning
 - Rocky cliffs
 - Produces electricity
-

SLIDE 5

Citizens Need Your Help

Visual:
Citizen cards

Lily (Age 8)

Needs:

- Safe travel to visit grandparents

Concern:

- Afraid of storms
-

Mr. Rivera (Farmer)

Needs:

- Deliver crops

Concern:

- Food spoils quickly
-

Nurse Maya

Needs:

- Deliver medicine

Concern:

- Emergencies cannot wait
-

Grandpa Ben

Needs:

- Reliable transportation

Concern:

- Cannot walk long distances
-

Delivery Team

Needs:

- Fast routes

Concern:

- Dangerous weather
-

Teacher Prompt:

Who do you think has the hardest challenge?

Why?

PHASE 2

Investigation & Research (10 Minutes)

SLIDE 6

Understanding The Problem

Students investigate:

What challenges does each island face?

What transportation options might work?

What could go wrong?

SLIDE 7

Transportation Research Board

Visual Cards:



Train



Bridge

 Cable Car

 Ferry

 Helicopter

 Tunnel

Students discuss:

Advantages

Disadvantages

Weather risks

Cost

SLIDE 8

New Restrictions Arrive

Teacher reveals:

Budget Limit

You may only build TWO major transportation systems.

Environmental Rule

No solution may damage wildlife habitats.

Safety Rule

Children and elderly citizens must travel safely.

Teacher Prompt:

Which ideas are no longer possible?

What must your team change?

PHASE 3

Collaborative Planning (10 Minutes)

SLIDE 9

Engineering Design Room

Using Excalidraw:

Students sketch:

- Island routes
 - Transportation connections
 - Emergency alternatives
-

SLIDE 10

Empathy Checkpoint

Before finalizing designs, teams answer:

For Children

How will they travel safely?

For Elderly Citizens

How will they travel comfortably?

For Workers

How will deliveries remain efficient?

For Emergencies

What happens during severe weather?

Teacher Challenge:

If you were Grandpa Ben, would you feel safe using your transportation system?

SLIDE 11

Community Council Vote

Students use reactions:

 Safe

 Helpful

 Fast

 Creative

Discuss improvements.

PHASE 4

Design Challenge (15 Minutes)

SLIDE 12

Build Your Solution

Teams create a Canva Transportation Proposal.

Must include:

- Transportation type
 - Route map
 - Safety features
 - Emergency plan
 - Who benefits most
-

SLIDE 13

Blueprint Creation

Students design:

Main Connection Route

Backup Route

Weather Protection Features

Examples:

- Heated bridge pathways
 - Covered cable cars
 - Lightning protection towers
 - Emergency rescue stations
-

SLIDE 14

Present Your Design

Teams explain:

1. What did you build?

2. Who does it help?
 3. Why is it safe?
 4. How does it work in different seasons?
-

PHASE 5

Twist Event (5 Minutes)

SLIDE 15

BREAKING NEWS

Visual:

Emergency weather alert

Teacher Announcement:

Attention Engineers!

A powerful blizzard has struck Winter Island.

Visibility is near zero.

Roads have frozen.

Travel conditions have become extremely dangerous.

SLIDE 16

Emergency Redesign

Students have 3 minutes.

Questions:

- What part of your system fails?
- How can people still travel?
- What backup plan can help citizens?

Possible solutions:

- Underground tunnels
 - Emergency shelters
 - Heated transport stations
 - Rescue vehicles
-

Teacher Prompt:

Great designers don't panic when plans fail.

They improve their designs.

PHASE 6

Reflection Debrief (5 Minutes)

SLIDE 17

Ark Reflection Circle

Empathy Reflection

1. Which citizen influenced your design most?
 2. Who benefited from your solution?
 3. Did anyone's needs conflict with another person's needs?
 4. How did your team handle that?
-

Adaptability Reflection

1. What changed after the blizzard?
 2. What part of your plan had to be redesigned?
 3. How did your team respond to the challenge?
-

Collaboration Reflection

1. Did everyone contribute ideas?
 2. How did your group solve disagreements?
 3. What was your team's strongest decision?
-

SLIDE 18

Mayor Aurora's Message

Visual:

The islands now connected

Dialogue:

Outstanding work, Engineers.

You did more than build transportation.

You rebuilt connections between people.

You considered different needs.

You adapted when conditions changed.

You showed what true problem-solvers do.

Mission Status:

CONNECTION RESTORED

Exit Ticket

Students type in chat:

Today I helped...

(A child / farmer / nurse / elderly citizen)

One thing I changed in my design was...

The biggest challenge my team solved was...

If I could improve my design tomorrow, I would...

Ark Value Connection

Empathy

Students learn that great solutions are not designed for themselves alone. They are designed by understanding the experiences, challenges, and needs of other people.

Adaptability

Students discover that real-world problems rarely stay the same. Successful innovators improve their plans when new information appears.

Growth Mindset

Students understand that redesigning is not failure—it is part of the innovation process.

Transition to Session 3

As the islands celebrate their reconnection, a strange signal is detected from deep beneath the ocean. An ancient underwater research station has suddenly come online after decades of silence...

Mission 003 awaits. 

Ark Academy US Pilot Program

Session 3: MYSTERY OF THE MISSING STUDENT

The Ark Challenge Files

Ages: 8–11

Duration: 50 Minutes

Students: 6

Core Values: Resilience + Empathy

Supporting Skills: Deductive Reasoning, Observation, Communication, Emotional Awareness

SESSION OVERVIEW

Big Idea

Students become members of the Ark Investigation Bureau after a student named Eli mysteriously disappears after school. What initially appears to be a simple missing-person case slowly unfolds into a complex puzzle filled with coded messages, misleading clues, emotional context, and hidden evidence.

Unlike Session 1, where students focused on solving a mystery, this mission challenges them to understand the emotions and experiences behind the mystery. Students must learn that good investigators do not jump to conclusions. They gather evidence, consider perspectives, remain calm during setbacks, and continue searching for answers even when things become confusing.

This session intentionally develops resilience by introducing incorrect assumptions, false leads, and contradictory evidence throughout the investigation.

FACILITATOR SETUP

Virtual Background

Ark Investigation Bureau Headquarters

Sound Effects

- School dismissal bell
- Police radio static
- Suspense music
- Notification alerts
- Investigation briefing sound

Materials Needed

- Google Meet
- Google Docs Detective Notebook
- Excalidraw Evidence Board
- Canva Investigation Report Template
- Evidence Cards
- Attendance Logs
- Student Notebook Pages
- School Map
- Secret Clue Cards

LEARNING TARGETS

Students Will:

- Analyze clues before making conclusions
- Demonstrate persistence during confusing situations
- Consider multiple explanations for evidence
- Understand emotions and perspectives within a situation
- Collaborate respectfully when opinions differ

CANVA STORYBOARD FLOW FOR DESIGNER

SLIDE 1

THE ARK CHALLENGE FILES

Mission 003:

Mystery of the Missing Student

Visual:

School building at sunset

Flashing investigation lights

Student backpack left behind

Text:

A student has disappeared.

The clues don't seem to make sense.

Can your team solve the mystery before nightfall?

SLIDE 2

Principal's Emergency Message

Visual:

School principal in office

Teacher reads dramatically:

Investigators...

We have a serious situation.

One of our students, Eli, never arrived home after school.

His family has called the school.

We found unusual markings on his papers and notebook.

We need your help finding him.

SLIDE 3

Missing Student Profile

Visual:

Student information card

Meet Eli

Age: 10

Favorite Subject:

Science

Favorite Activity:

Exploring and building things

Personality:

Curious, creative, adventurous

Known Facts:

✓ Attended school today

✓ Left after dismissal

✓ Did not arrive home

✓ Left several clues behind

Teacher Prompt:

What do we know?

What don't we know yet?

PHASE 1

Mission Briefing (5 Minutes)

SLIDE 4

Evidence Packet #1

Students receive:

-  Attendance Record
-  Homework Sheet
-  Science Quiz
-  Notebook Page
-  School Map

Teacher explains:

Everything matters.

Some clues are useful.

Some clues are distractions.

Good investigators must determine the difference.

SLIDE 5

First Clue

Visual:

Science Quiz

Top corner contains strange letters:

L

I

B

R

A

R

Y

Teacher asks:

Why would these letters appear separately?

Students notice:

LIBRARY

Possible lead discovered.

PHASE 2

Evidence Analysis (10 Minutes)

SLIDE 6

Detective Notebook

Google Docs Shared Investigation Sheet

Students organize:

Evidence	Observation	Possible Meaning
----------	-------------	------------------

Quiz Paper		
------------	--	--

Notebook		
----------	--	--

Attendance Log		
----------------	--	--

School Map		
------------	--	--

SLIDE 7

Notebook Discovery

Visual:

Notebook page

Text:

Sometimes the quietest place has the loudest answers.

Students discuss:

What location might this describe?

Possible answers:

- Library
 - Reading Room
 - Study Hall
-

SLIDE 8

School Map Analysis

Visual:

Map of school

Locations marked:

Library

Gym

Science Lab

Garden

Storage Room

One location circled lightly in pencil.

Students discover:

Science Lab

SLIDE 9

Emotional Evidence

Visual:

Journal entry

Excerpt:

I hope nobody thinks I'm in trouble.

I just need to finish something important before tomorrow.

Teacher Prompt:

What emotions might Eli be feeling?

Worried?

Excited?

Nervous?

Determined?

Empathy Discussion:

Understanding feelings can help solve mysteries.

PHASE 3

Investigation Planning (10 Minutes)

SLIDE 10

Detective Team Roles

Students choose:

 Evidence Analyst

 Mapper

 Recorder

 Clue Connector

 Lead Presenter

 Logic Checker

SLIDE 11

Secret Intelligence Reports

Teacher privately messages clues.

Examples:

Student A receives:

Eli borrowed a science book yesterday.

Student B receives:

A teacher saw Eli carrying a flashlight.

Student C receives:

The science lab was open after school.

Students must communicate effectively to combine information.

SLIDE 12

Theory Board

Teams discuss:

Possible Location

Possible Reason

Supporting Evidence

Questions Remaining

Teacher Prompt:

What evidence supports your theory?

What evidence challenges it?

PHASE 4

Final Deduction Challenge (15 Minutes)

SLIDE 13

Build Your Evidence Board

Using Excalidraw

Students organize:

Evidence

Connections

Theories

Final Conclusion

Visual arrows connect clues together.

SLIDE 14

The Hidden Code

Notebook clue:

Every first letter spells:

S C I E N C E

Students realize:

Science is central to the mystery.

SLIDE 15

Final Discovery

Students determine:

Eli's Location

Old Science Greenhouse

Why Eli Disappeared

Eli was secretly finishing a science project for the school showcase.

He lost track of time and his phone battery died.

He was never in danger.

Rescue Plan

Contact staff

Follow map clues

Travel safely to greenhouse

Inform family

Teacher Prompt:

How do we know this conclusion makes sense?

Students present evidence.

PHASE 5

Twist Event (5 Minutes)

SLIDE 16

NEW EVIDENCE FOUND

Visual:

Investigation Alert

Teacher announces:

Detectives!

One piece of evidence was intentionally planted as a fake clue.

Reveal:

The library clue was false.

Students must reassess.

SLIDE 17

Reopen The Case

Questions:

What evidence still supports your conclusion?

What evidence should be removed?

Does your conclusion still make sense?

Can you prove it?

Teacher emphasizes:

Strong thinkers revise ideas when new information appears.

Resilience Moment

Students experience:

- Incorrect assumptions
- Frustration
- Uncertainty

Goal:

Practice continuing despite mistakes.

PHASE 6

Reflection Debrief (5 Minutes)

SLIDE 18

Ark Reflection Circle

Resilience Reflection

1. How did your team react when a clue turned out to be wrong?
 2. What helped you keep going?
 3. What did you do when your first idea didn't work?
-

Empathy Reflection

1. How do you think Eli felt during the situation?
 2. How did understanding his feelings help solve the mystery?
 3. Why is it important not to assume the worst about people?
-

Adaptability Reflection

1. What changed after the fake clue was revealed?
2. How did your team adjust?
3. What evidence became most important?

Collaboration Reflection

1. How did your team solve disagreements?
 2. Did everyone contribute?
 3. What made your teamwork successful?
-

SLIDE 19

Case Closed

Visual:

Greenhouse at sunset

Eli waving

Principal speaking

Dialogue:

Outstanding work, Investigators.

You followed evidence carefully.

You stayed calm when things became confusing.

You reconsidered your assumptions.

Most importantly, you remembered that every mystery involves real people and real feelings.

Mission Status:

CASE SOLVED

Exit Ticket

Type in chat:

One clue that fooled me was...

One way I showed resilience today was...

One feeling Eli may have experienced was...

One thing good investigators should always remember is...

Ark Value Connection

Resilience

Students learn that mistakes, confusion, and dead ends are normal parts of solving difficult problems. Success comes from continuing to think, revise, and investigate.

Empathy

Students discover that understanding a person's emotions and motivations can be just as important as analyzing physical evidence.

Adaptability

Students practice changing their conclusions when new information emerges rather than becoming attached to being right.

Transition to Session 4

As Eli is safely reunited with his family, investigators discover something unexpected inside the greenhouse: a locked metal case containing blueprints, coded messages, and a mysterious note.

The note reads:

"If you found this, you're ready for the final challenge."

Mission 004: The Future City Crisis awaits. 🚀🏙️

Ark Academy US Pilot Program

Session 4: NO ONE GETS LEFT BEHIND

The Ark Challenge Files – Final Mission

Ages: 8–11

Duration: 50 Minutes

Students: 6

Core Values: Empathy + Resilience + Adaptability + Growth Mindset

Supporting Skills: Ethical Decision-Making, Resource Management, Leadership, Negotiation, Engineering Design

SESSION OVERVIEW

Big Idea

The world is facing its greatest crisis.

A sudden atmospheric collapse has made Earth increasingly unsafe for human life. Scientists have identified a temporary safe colony on a nearby planet, but evacuation ships are limited, resources are scarce, and launch windows are closing rapidly.

Students become members of the **Ark Evacuation Council**, humanity's final emergency planning team.

Their challenge is not simply to build a spaceship.

Their challenge is to build a system that protects everyone.

The most important rule:

NO ONE GETS LEFT BEHIND.

Students must balance engineering, ethics, fairness, empathy, and problem-solving while working under increasing pressure and changing conditions.

This final mission intentionally combines all four Ark Core Values learned throughout the pilot:

- Growth Mindset
 - Resilience
 - Empathy
 - Adaptability
-

FACILITATOR SETUP

Virtual Background

Ark Space Command Center

Sound Effects

- Mission control announcements
 - Rocket launch countdown
 - Emergency warning alerts
 - Radio transmissions
 - Atmospheric storm sounds
 - Final launch sequence
-

Materials Needed

- Google Meet
 - Excalidraw Rocket Planning Board
 - Canva Blueprint Template
 - Passenger Cards
 - Resource Cards
 - Emergency Event Cards
 - Shared Google Doc Planning Sheet
-

LEARNING TARGETS

Students Will:

- Make ethical decisions that prioritize fairness and inclusion
 - Balance limited resources while meeting community needs
 - Revise plans under pressure
 - Collaborate effectively during difficult decisions
 - Justify choices using evidence and reasoning
-

STORY CONNECTION

This mission begins immediately after Session 3.

When Eli's mysterious metal case was opened, investigators discovered hidden files from the future.

Inside was a warning:

EARTH ATMOSPHERIC COLLAPSE DETECTED.

EVACUATION PROTOCOL REQUIRED.

HUMANITY SURVIVAL DEPENDS ON COLLABORATIVE LEADERSHIP.

The Ark Investigation Bureau immediately activates its highest-level emergency response team:

The students.

CANVA STORYBOARD FLOW FOR DESIGNER

SLIDE 1

THE ARK CHALLENGE FILES

FINAL MISSION

NO ONE GETS LEFT BEHIND

Visual:

Earth surrounded by storms

Large evacuation rocket preparing for launch

Countdown clock

Text:

Atmospheric Failure Detected

Evacuation Window:

40 Minutes Remaining

SLIDE 2

Emergency Transmission

Visual:

Static-filled transmission

Teacher reads dramatically:

Attention Ark Council.

Earth's atmosphere is becoming unstable.

Survival colonies are ready.

However...

Transportation capacity is limited.

Resources are running low.

Every decision matters.

Humanity is depending on you.

SLIDE 3

Mission Objective

Visual:

Mission dashboard

Students must:

- ✓ Design an evacuation rocket
 - ✓ Fit every passenger safely
 - ✓ Manage limited resources
 - ✓ Adapt to emergencies
 - ✓ Ensure nobody is excluded
-

PHASE 1

Mission Briefing (5 Minutes)

SLIDE 4

Passenger Manifest

Visual:

Passenger Cards

 Grandma Rosa

Needs:

- Mobility support

 Twin Children

Needs:

- Adult supervision

 Injured Scientist

Needs:

- Medical station

 Engineer

Needs:

- Access to control systems

 Service Dog

Needs:

- Safe accommodation

 Doctor

Needs:

- Medical supplies

Teacher Prompt:

Does everyone need exactly the same things?

How can fairness sometimes look different from equality?

SLIDE 5

Available Resources

Visual:

Resource inventory

Students receive:

Oxygen Tanks = 10

Food Units = 12

Sleeping Pods = 8

Medical Kits = 3

Energy Cells = 5

Space Capacity = Limited

Teacher explains:

You have enough resources to survive.

But only if you use them wisely.

PHASE 2

Resource Analysis (10 Minutes)

SLIDE 6

Engineering Challenge Board

Students discuss:

What resources are most important?

What can be shared?

What cannot be shared?

How do we ensure everyone is safe?

SLIDE 7

Community Needs Analysis

Students complete chart:

Passenger	Needs	Priority Considerations
Grandma Rosa		
Children		
Scientist		
Doctor		

Service Dog

Teacher Prompt:

Who might be affected most by poor planning?

SLIDE 8

Ethical Decision Point

Teacher introduces scenario:

You only have 8 sleeping pods.

There are 10 passengers.

Nobody may be excluded.

What will your team do?

Possible solutions:

- Shared sleeping schedules
- Multi-use spaces
- Foldable sleeping systems

Purpose:

Move students away from elimination thinking.

PHASE 3

Collaborative Engineering (10 Minutes)

SLIDE 9

Rocket Design Studio

Using Excalidraw

Students sketch:

- Sleeping quarters
 - Medical station
 - Food storage
 - Oxygen systems
 - Control room
 - Recreation area
-

SLIDE 10

Leadership Challenge

Teacher Prompt:

Not everyone will agree.

Strong leaders don't control conversations.

They help everyone contribute.

Students practice:

- Listening
 - Negotiating
 - Building on ideas
-

SLIDE 11

Inclusion Checkpoint

Before finalizing:

Teams answer:

- ✓ Can every passenger travel safely?
 - ✓ Are unique needs addressed?
 - ✓ Is anyone unintentionally forgotten?
 - ✓ What happens during emergencies?
-

PHASE 4

Rocket Design Challenge (15 Minutes)

SLIDE 12

Build The Ark Ship

Students create Canva blueprint including:

Rocket Layout

Resource Plan

Safety Systems

Emergency Protocols

Passenger Accommodations

SLIDE 13

Survival Strategy

Students explain:

How oxygen is managed

How food is distributed

How emergencies are handled

How fairness is maintained

SLIDE 14

Council Presentation

Teams present:

1. What did you design?
 2. How does it work?
 3. How does it help everyone?
 4. Why is it fair?
-

Teacher asks:

What sacrifices did your team have to make?

PHASE 5

Twist Event (5 Minutes)

SLIDE 15

INCOMING EMERGENCY

Visual:

Red flashing warning screen

Teacher Announcement:

Attention Ark Council.

Two additional survivors have arrived at the launch site.

Launch is in 5 minutes.

There is no second rocket.

Visual:

 Child Survivor

 Farmer Survivor

SLIDE 16

Emergency Redesign

Students have 3 minutes.

Questions:

How will you fit everyone?

What resources must change?

How can you stay fair?

What new compromises are needed?

Teacher Prompt:

Fairness becomes most important when resources become scarce.

SLIDE 17

Final Vote

Poll Question:

Which solution best protects everyone?

Students must justify decisions.

PHASE 6

Reflection Debrief (5 Minutes)

SLIDE 18

Ark Reflection Circle

Empathy Reflection

1. How did your team make sure everyone was considered?
 2. Which passenger's needs were hardest to solve?
 3. Why is inclusion important during emergencies?
-

Resilience Reflection

1. What challenge frustrated your team most?
 2. How did your team keep working?
 3. What helped you continue despite pressure?
-

Adaptability Reflection

1. What changed after the new survivors arrived?
2. What redesign was necessary?
3. How did your team adjust?

Growth Mindset Reflection

1. What idea improved because of feedback?
 2. What mistake helped your team learn?
 3. What would you redesign if you had more time?
-

SLIDE 19

Final Mission Success

Visual:

Rocket launching from Earth

Students looking out spacecraft windows

Countdown reaching zero

Mission Control Message:

Ark Council,

You have completed all four challenge files.

You solved mysteries.

You rebuilt communities.

You designed solutions.

You helped people.

Most importantly...

You proved that great leaders make sure no one gets left behind.

Mission Status:

HUMANITY SAVED

SLIDE 20

THE ARK VALUES REVEALED

Visual:

Four interconnected symbols

Growth Mindset

Great thinkers improve ideas.

Resilience

Great thinkers keep going.

Empathy

Great thinkers care about people.

Adaptability

Great thinkers adjust when circumstances change.

Final Message

These weren't just mission skills.

These are life skills.

The best innovators, leaders, and problem-solvers use them every day.

Final Exit Ticket

Students type:

The most important thing I learned about leadership is...

One way I showed empathy today was...

One challenge I overcame was...

One Ark value I will continue using outside class is...

My Ark Mission Rating



PROGRAM CULMINATION

At the end of Session 4, students have experienced a complete progression of developmental challenges:

Session 1

Escape the Mall

Growth Mindset + Adaptability

"Problems can be solved in many ways."

Session 2

The Four-Season Islands

Empathy + Adaptability

"Good solutions consider everyone's needs."

Session 3

Mystery of the Missing Student

Resilience + Empathy

"Understanding people is as important as understanding facts."

Session 4

No One Gets Left Behind

Empathy + Resilience + Growth Mindset + Adaptability

"The best leaders solve problems without leaving anyone behind."

This final mission serves as the capstone experience where students apply every Ark Core Value in a high-stakes, collaborative challenge that emphasizes ethical leadership, inclusion, and human-centered problem-solving.

Student - Exit Ticket

STUDENT SELF-EVALUATION EXIT TICKET

To be completed at the end of every session.

Reflection Questions

Growth Mindset

1. What challenge was hardest for you today?
2. What strategy did you try when something did not work?

Collaboration

3. How did you help your team today?
4. How did your teammates help you?

Adaptability

5. What changed during the mission that forced your team to adjust?
6. How did you react when the plan changed?

Emotional Reflection

7. What moment made you feel proud today?
8. What is one thing you would improve next time?

Self-Rating Scale

Students rate themselves from 1-4:

Skill	Self Score
I contributed ideas to my team	
I listened to others respectfully	
I stayed focused during challenges	

I kept trying even when things were
difficult

I adapted when plans changed

Do you have any feedback regarding your class?

Teacher After Class Form

AFTER-CLASS FACILITATOR OBSERVATION FORM

CLASS INFORMATION

Session Number: ☐ Session 1 ☐ Session 2 ☐ Session 3 ☐ Session 4

Mission Title:

Date:

Facilitator:

Class/Group:

Number of Students Present:

PART 1. SESSION REFLECTION

Overall Class Energy

- ☐ Very Engaged
- ☐ Mostly Engaged
- ☐ Mixed Engagement
- ☐ Low Engagement

Difficulty Level

- ☐ Too Easy
- ☐ Appropriate Challenge
- ☐ Difficult but Productive
- ☐ Too Difficult

How successful was today's mission overall?

1 ☐ Poor

2 ☐ Fair

3 ☐ Good

4 ☐ Excellent

Comments:

PART 2. CORE VALUE OBSERVATIONS

Rate overall class performance.

Domain	1	2	3	4
Growth Mindset	☐	☐	☐	☐
Resilience	☐	☐	☐	☐
Empathy & Social Awareness	☐	☐	☐	☐
Adaptability	☐	☐	☐	☐
Collaborative Communication	☐	☐	☐	☐
Critical Thinking & Problem Solving	☐	☐	☐	☐
Creative & Innovative Thinking	☐	☐	☐	☐
Digital Collaboration	☐	☐	☐	☐

PART 3. SESSION-SPECIFIC OBJECTIVES

Session 1: Escape the Mall

- ☐ Students tried multiple strategies.
- ☐ Students adjusted when new information appeared.
- ☐ Students communicated effectively with teammates.
- ☐ Students remained engaged despite mistakes.

Comments:

Session 2: Four-Season Islands

- ☐ Students considered the needs of others.
- ☐ Students redesigned solutions after changes.
- ☐ Students balanced creativity with practicality.
- ☐ Students collaborated respectfully during disagreements.

Comments:

Session 3: Mystery of the Missing Student

- ☐ Students persisted despite confusion.
- ☐ Students analyzed evidence logically.
- ☐ Students recovered after incorrect assumptions.
- ☐ Students solved disagreements respectfully.

Comments:

Session 4: No One Gets Left Behind

- ☐ Students considered fairness and inclusion.
- ☐ Students revised plans under pressure.
- ☐ Students justified decisions logically.
- ☐ Students ensured everyone participated.

Comments:

PART 4. OBSERVABLE BEHAVIORS

Growth Mindset

Observed examples:

- ☐ Tried different approaches
- ☐ Accepted feedback positively
- ☐ Learned from mistakes
- ☐ Took intellectual risks
- ☐ Encouraged teammates

Notes:

Resilience

Observed examples:

- ☐ Stayed engaged during setbacks
- ☐ Recovered after failure
- ☐ Managed emotions productively
- ☐ Continued participating despite confusion

Notes:

Empathy

Observed examples:

- ☐ Included quieter students
- ☐ Listened respectfully
- ☐ Supported peers
- ☐ Demonstrated fairness

Notes:

Adaptability

Observed examples:

- ☐ Revised plans quickly
- ☐ Accepted unexpected changes
- ☐ Shifted strategies effectively
- ☐ Remained productive under uncertainty

Notes:

PART 5. GROUP DYNAMICS

Team Collaboration

- ☐ Excellent
- ☐ Good
- ☐ Developing
- ☐ Needs Support

Discussion Quality

- ☐ Highly Productive
- ☐ Productive
- ☐ Inconsistent
- ☐ Limited

Participation Distribution

- ☐ Everyone contributed
- ☐ Most contributed
- ☐ Few students dominated
- ☐ Significant imbalance

Conflict Resolution

- ☐ Independent
- ☐ Needed occasional support
- ☐ Needed frequent support

Comments:

PART 6. INDIVIDUAL STUDENT NOTES

Students Showing Strong Growth

Student Name	Evidence Observed
-------------------------	--------------------------

_____	_____
_____	_____
_____	_____

Students Needing Additional Support

Student Name	Area for Support	Notes
_____	_____	_____
_____	_____	_____
_____	_____	_____

PART 7. NOTABLE MOMENTS

Leadership Moments

Who demonstrated leadership?

Evidence:

Creative Solutions

Interesting ideas students generated:

Memorable Quotes or Conversations

Breakthrough Moments

PART 8. FACILITATOR SUMMARY

Biggest Strengths Observed Today

- ☐ Growth Mindset
- ☐ Resilience
- ☐ Empathy
- ☐ Adaptability
- ☐ Collaboration
- ☐ Critical Thinking
- ☐ Creativity
- ☐ Digital Collaboration

Examples:

Areas for Continued Development

- ☐ Growth Mindset
- ☐ Resilience

- ☐ Empathy
- ☐ Adaptability
- ☐ Collaboration
- ☐ Critical Thinking
- ☐ Creativity
- ☐ Digital Collaboration

Examples:

PART 9. PREPARATION FOR NEXT SESSION

Students may need support with:

- ☐ Collaboration
- ☐ Communication
- ☐ Emotional Regulation
- ☐ Logical Reasoning
- ☐ Adaptability
- ☐ Leadership
- ☐ Inclusion
- ☐ Confidence

Specific Notes:

OVERALL FACILITATOR REFLECTION

What surprised you today?

What worked particularly well?

What would you modify next time?

One sentence describing today's class:

Overall Session Rating



1 ☐ Emerging

2 ☐ Developing

3 ☐ Proficient

4 ☐ Advanced